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POWER SYSTEM STABILITY AND CONTROL COMPARATIVE ANALYSIS

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Abstract: We identify different types of power system stability and find that transient stability is the one which imposes the most stringent requirements. On the other hand, we identify the three classes of approaches to transient stability assessment and examine whether and which ones are able to meet such requirements. It is found that this may be achieved by appropriate combinations, lying at the crossroads of numerical, direct, and automatic learning methods. It is also found that many of the combinations advocated for the purpose of transient stability are readily adaptable to other stability problems. Finally, a general approach is anticipated where all types of stability would be handled in a unified way.

Keywords: Power systems; stability analysis; transient stability assessment; time-domain methods; artificial intelligence; decision trees; neural networks.

1. INTRODUCTION
POWER SYSTEM STABILITY

Power system stability may broadly be defined as that property of a power system that enables it to remain in a stable equilibrium state under normal operating conditions and to regain an acceptable equilibrium state after being subjected to a disturbance (Kundur, 1994). Of all complex power system phenomena, stability is the most intricate to understand and to handle.

Power system stability is a single yet multifaceted problem depending upon a variety of factors, such as : the time span that must be taken into consideration in order to assess (in)stability; the size of the disturbance considered; the physical nature of the resulting instability. Hence, although the rigorous classification among distinct types of stabilities is difficult, a non-rigorous classification often accepted for practical use may be established, relying on the above factors. Thus, with reference to the time span of the phenomena, one distinguishes short-term from mid-term and long-term stability. With reference to the size of the disturbance considered, one distinguishes small-disturbance from large-disturbance stability : the former may be handled via linearization of the dynamic equations of motion, the latter requires nonlinear approaches. Further, both the small-disturbance and large-disturbance stability phenomena may be subdivided into "voltage" (i.e. voltage magnitude) and "angle" (i.e. voltage phase angle) ones. Table 1 illustrates these latter classifications. In particular, it indicates that the angle large-disturbance stability is the so-called transient stability.

Table 1. Classification of voltage and angle stabilities

	Stability in the small Linearized analysis	Stability in the large Nonlinear analysis
Voltage Stability	Small-disturbance	Large-disturbance
Angle Stability	Small-disturbance	Large-disturbance i.e. Transient

Transient stability of a power system is its ability to maintain machines' synchronous operation when subjected to a large disturbance. At the occurrence of such a disturbance, the system response involves large excursions of machine rotor angles and, whenever corrective actions fail, loss of synchronism among machines. Generally, the loss of synchronism develops in a very few seconds after the disturbance inception; actually, among the phenomena listed in Table 1 transient stability is the fastest to develop.

The nonlinear character of transient instability, its fast nature and its disastrous practical implications make it one of the most important and at the same time most problematic issues to analyse and even more to control, especially in real-time. These challenging aspects motivate our choice to mainly concentrate on transient stability.

The purpose of this paper is to present a synthetic view of approaches to transient stability assessment. It

describes the three classes of methods available nowadays and points out their features and complementary strengths. It further suggests how to choose these latter in order to meet specific requirements of power system planning and operation. It also shows that some of the methods developed in the context of transient stability can readily be adapted to other types of stability, in particular to voltage stability. The paper finishes up with the perspective of a unified approach able to handle simultaneously all types of stability.

Remarks

1.- Large-distubance voltage stability is yet another challenging issue of great practical importance, though less fast to develop than transient instability (typically some (tens of) minutes vs very few seconds for the latter). In addition to the reasons advocated above, we chose to deal with transient stability because it is richer from a methodological point of view : it has one more class of sound approaches than voltage stability namely, the class of direct methods[1].

2.- The present paper is meant as a tutorial, not as a critical survey. Hence, the references do not properly reflect the huge amount of research effort developed during the last three decades. Comprehensive lists of references may be found in dedicated books such as (Pai, 1981; Pai, 1989; Fouad and Vittal, 1992; Pavella and Murthy, 1993).

2. TRANSIENT STABILITY APPLICATION CONTEXTS - NEEDS

2.1 *Transient stability An introductory intuitive description*

Broadly speaking, transient stability entails the evaluation of a power system's ability to withstand large disturbances and to survive transition to a normal or acceptable operating condition. "Large disturbances" are meant to be severe faults such as short-circuit on a transmission line, loss of generators/loads, loss of a portion of transmission network. The transients initiated by such disburbances are generally electromechanical in nature; their behaviour is fairly complex and cannot be predicted easily. They differ from one power system to another and from one situation to another. They also largely depend on the origin, type, location and severity of disturbance. Hence large number of simulations are carried out regularly and during planning stages to gain knowledge of the system. Yet, even a well designed and normally operated system may face the threat of transient stability. Hence, with the growing stress on today's power systems which are forced to operate closer to their stability limits, there is a pressing need for on-line transient stability assessment.

Transient stability is almost an as old issue as power systems themselves. Its classical definition proposed in 1948 (Kimbark, 1948) is as follows.

> "Power system (transient) stability is a term applied to alternating current electric power systems, denoting a condition in which the various synchronous machines of the system remain in *"synchronism"*, or *"in step"* with each other. Conversely, instability denotes a condition involving *"loss of synchronism"*, or falling *"out of step"*."

Hence, for a simplified intuitive description of transient stability, a power system may be regarded as a set of synchronous machines and of loads interconnected through the transmission network. Under normal operating conditions, all the system machines run at the synchronous speed. If a large disturbance occurs the machines start "swinging" with respect to each other, their motion being governed by nonlinear differential equations. Their number depends on the power system modelling; it may be orders of magnitude larger than the number of system machines.

From a system theory point of view transient stability is a strongly nonlinear, highly dimensional problem.

2.2 *Transient stability limits*

The conventional transient stability measure of the system robustness to withstand a large disturbance is its corresponding critical clearing time (CCT) : this is the maximum time duration that the disturbance may act without the system losing its capability to recover a steady-state (i.e. stable) operation[2].

Another transient stability measure of great practical importance is the power (generation or transfer) limit. It is defined as the largest power sustainable without loss of synchronism, given the occurrence of a large disturbance and its clearing scenario (i.e. the sequence of actions to eliminate it).

Note that the above measures imply consideration of three distinct phases : the pre-fault (i.e. pre-disturbance), the during-fault, and the post-fault one.

2.3 *Preventive-mode application contexts Corresponding needs*

The diversity in power system morphology and operation strategies induces various needs. However, despite their specifics, all power systems have also common needs, relative to the particular field of application. In the preventive mode, we traditionally distinguish three such fields, each with its own requirements. They are summarized in Table 2 and briefly commented below.

Power system planning. In this context, multitudinous cases must be performed, months to years before the planned system is finally designed.

Despite the long time allotted to these simulations, speed is yet an important factor. Indeed, to overcome the "curse of dimensionality", a rapid, screening tool is needed to identify the "interesting" situations on which

[1] Despite some valuable attempts, direct methods have not as yet proven to be practically effective for voltage stability assessment.

[2] Typically, the "interesting" range of CCT values is in between 0 and some 200-300 ms. CCTs larger than, say, 400 ms are considered to correspond to "uninterestingly mild" contingencies.

Table 2. Preventive-mode application contexts and their needs. Adapted from (Pavella and Murthy, 1993).

Application (time allotted)	Key requirements		Key features	Needs
	Speed	Accuracy		
System planning (months to decades)	Desired	Important	Voluminous cases	• Screening tools • Sensitivity
Operation planning (days to hours)	Critical	Important	Combining economy with security	• Margins • Sensitivity
Real-time operation (few minutes)	Crucial	Desired	Time is crucial	Real-time (*) • Analysis • Sensitivity • Control

(*) Lack of appropriate techniques may lead to operation stability margins which are needlessly large and anti-economic

the planner should concentrate. On the other hand, a sensitivity assessment method is desired, able to provide information about the relevant system parameters, their impact on stability, and thence to suggest means to reinforce stability, whenever needed (Kundur *et al.*, 1987).

Operation planning. In these studies, where the time horizon reduces to days or hours, the speed becomes critical. Moreover, as the power system is operated in ways not necessarily envisioned during the system design, there is a need, in addition to the crude critical clearing time assessment to appraise stability margins and, whenever necessary, to suggest means to increase them. Once again, sensitivity analysis and means to control, in addition to analysis are the essential characteristics required. They need not be very accurate, provided that they allow the operator to make decisions with fewer stability runs.

In the present-day practices where operation planning stability studies merely reduce to the form of a stability limit or operating guide, techniques based on oversimplified modelling are acceptable : they may be used to identify "interesting" cases ("some answer is better than no answer at all") and possibly to suggest corrective actions.

Real-time operation. Here, only some few minutes are left to the operator to both analyse the situation and if necessary make decision : hence, speed becomes a crucial factor; besides, sensitivity and control tools are strongly desired.

2.4 *Emergency mode*

Table 2 gathers the salient features and needs of the above three application contexts in the *preventive mode*. The concern of this mode is to forecast the projected situation, though in very different time horizons, so as to best assess the system transient stability limits which ensure its safe operation. Or, to predetermine appropriate corrective actions which should be taken to ensure stability, if some severe disturbance occurs.

However, even in the context of real-time operation, it is very unlikely that stability limits can be optimised, because of the combinatorial nature of the events which might occur. Ideally, the final control decision should be taken for the *actual* system state, after a (large) disturbance has *actually* occurred. This should be the task of the real-time emergency transient stability assessment and control.

However, such a task is very demanding, given the extremely short times left to take actions in order to preserve system's integrity. Yet, nowadays, the need for real-time emergency control is strongly felt. In the coming sections we will suggest means to meet this need.

3. THE THREE CLASSES OF METHODS COMBINATIONS – HYBRIDISATIONS

There are three classes of approaches to transient stability assessment (TSA) : time-domain, direct and automatic learning methods. They may be subdivided into conventional and non-conventional, or into deterministic and non-deterministic (statistical) ones. Table 3 illustrates these two types of classification. Below we briefly describe them and point out their salient features.

Table 3. Classification of TSA methods

	Conventional	Non-conventional	
	Time-domain	Direct	Autom. learn.
Deterministic	x	x	
Statistical			x

3.1 *Time-domain (T-D) methods*

Transient stability phenomena are governed by two sets of equations :

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}) \quad (1)$$

$$\mathbf{0} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y}) \,. \quad (2)$$

Set (1) consists of differential equations of all machines, turbines and all types of controls; set (2) comprises the machine structure equations, equations of network and loads, etc. The dimensions of vector $\boldsymbol{x}$ depends on the modelling detail; it is lower bounded by twice the number of system machines, but may be orders of magnitude larger.

Upon the inception of a disturbance, time-domain methods solve numerically ("step-by-step") the above nonlinear sets of algebraic and differential equations, in both the during-fault and post-fault configurations of the power system. Usually, to search the corresponding transient stability limit (for example CCT) they proceed by dichotomy. The number of required transient stability runs depends on the accuracy sought. To fix ideas, a CCT search typically requires about 7 stability runs (for example, 3 stable and 4 unstable

ones)[3]. It amounts to about 15 minutes of CPU time on a Sparc 20 station, for a medium-size power system (comprising about 60 machines).

Historically, time-domain methods have been used before the advent of computers : calculations of very simplified versions of eqs. (1) and (2) were carried out manually to compute the "swing curves" of the machines, i.e. the machine rotor angles as functions of time (Park, 1929). Since, T-D methods have prevailed as the universal way of assessing transient stability of general multimachine power systems. This is owing to their intrinsic attractive features but also to the fact that, until recently, no general alternative approaches were available. At present, the conditions are changing : new methods start being effective, while at the same time more stringent requirements appear that T-D methods are unable to meet.

Below we stress upon the main pros and cons of T-D methods; they yield a twofold, apparently contradictory observation : T-D methods are definitely insufficient, yet indispensable to in-depth transient stability investigations.

Pros. T-D methods are able to :

- provide essential information about relevant parameters of the system evolution with time (machine swing curves, i.e. rotor angles; speeds; accelerations; powers; etc.);
- embed any power system modelling and stability scenario;
- reach the required accuracy, provided that the modelling of a power system is correctly designed and its parameters accurately known.

Hence, T-D methods are the reference for transient stability analysis. Besides, they are an essential auxiliary tool of non-conventional methods; this will become clear below.

Cons. T-D methods are unable to provide :

- straightforward screening tools in order to discard "uninterestingly harmless" disturbances;
- sound stability margins which would inform about "how far from (in)stability the system is, and which would yield suitable sensitivity analysis tools;
- sound suggestions to control.

In short, T-D methods cannot meet most of the needs identified in Table 2. Besides, they are totally unable to meet the needs of real-time emergency control.

[3]A stability run is meant to be the time interval during which the system equations (1), (2) are solved numerically. An "unstable run" starts at the disturbance inception and stops as soon as the machines irrevocably lose synchronism (e.g. as soon as the rotor angular difference of any two machines exceeds, say, 180°); typically, an unstable run may last for about 1 second after the disturbance clearance (which last for about 100 to 400 ms after the disturbance inception); but it may be much shorter.
On the other hand, a "stable run" covers the entire time interval deemed necessary to decide that the system is, indeed, stable; typically, this interval is of about 5 to 10 seconds; but it may be chosen shorter or larger depending upon the power system modelling details (oversimplified system modelling would not justify more than, say, 3 s interval) and the very nature of the involved physical phenomena (plant mode; inter-area mode; first- or multi-swing instabilities).

3.2 *Non-conventional approaches*
Historical background

The above deficiencies of T-D methods gave an impetus to the development of non-conventional approaches. They are of two types : direct (Lyapunov-like) and automatic learning ones. They both started being developed almost simultaneously, by the late sixties. Quite soon, however, it became clear that automatic learning methods could not develop satisfactorily because the computer facilities available at that moment were insufficient to meet their demanding needs. Thus, despite some thoughtful attempts, automatic learning started becoming effective only two decades later.

In the meantime, direct methods were receiving intensive research effort. One main attractiveness was their capability of reducing the T-D simulations to the sole during-fault period while avoiding all repetitive runs, i.e. of reducing simulations to a very small percentage of the overall computing effort of T-D methods. The other attractiveness is that direct methods may provide sound stability margins which in turn open possibilities towards suitable sensitivity analysis.

The above features of paramount practical importance, together with the fascination for the application of direct methods to power system transient stability explain the researchers' intensive efforts developed for almost three decades. Quite soon, however, the first enthusiastic expectations put on these methods have been overshadowed by two main difficulties. The one is linked to the impossibility of designing good Lyapunov functions for multimachine power systems, unless (over)simplified modelling is used. The other difficulty is relative to the assessment of a practical stability domain, suitable from both viewpoints: computational efficiency and accuracy of the transient stability assessment in the large. As often happens in applications, these difficulties have eventually been circumvented by combining theoretically pure approaches with pragmatic engineering solutions. And, as usually, many combinations have been set up, corresponding to various tradeoffs. They are shortly described below, in § 3.3. Anticipating, we only mention here that they mainly rely on direct criteria coupled with T-D simulations, and that, eventually, their achievements meet to a large extent the high expectations initially put on direct methods.

As for the automatic learning approaches, the first attempts to apply them to transient stability date back to the early 70's (Dy-Liacco, 1968; Prabhakara and El-Abiad, 1975; Prabhakara and Heydt, 1987). After a latent period of about two decades, they started actively being developed around the mid eighties. Beginning with ANN approaches (e.g., see (Sobajic and Pao, 1989)), they have gradually turned to machine learning methods, yielding decision trees (Wehenkel *et al.*, 1989), further complemented with k-NN methods (Houben *et al.*, 1995). By now, a whole bunch of automatic learning methods, often hybridised are available. They are described in § 3.4.

3.3 *Direct methods of the Lyapunov type*

Principle. The principle of the Lyapunov criterion applied to power system transient stability may be stated as follows : under preassigned stability conditions (i.e. preassigned disturbance and its clearing scenario) assess whether the system entering its post-fault configuration is stable, using the Lyapunov criterion. In turn, this yields the following practical procedure : upon constructing a Lyapunov function, compute its value $V(\boldsymbol{x}(t_e))$ and decide that the system is stable if this is smaller than $V_\ell = V(\boldsymbol{x}^u)$, unstable otherwise. Here, $\boldsymbol{x}(t_e)$ denotes the values taken by the components of the system state vector at t_e , the disturbance clearing time; these values are computed via T-D simulations carried out in the during-fault period up to t_e . On the other hand, $\boldsymbol{x}^u$ denotes the values that the components of the system state vector take on the boundary of the stability domain, or of its estimate.

$V = V_\ell$ provides the stability limit condition, i.e. the CCT for a given power level, or the power limit for a given t_e . On the other hand, the excess of V_ℓ over V represents a sound stability margin, measuring "how far from instability" the system is under the preassigned conditions.

From a computational point of view, the above procedure requires :

- computation of $V(\boldsymbol{x}(t_e))$ which is almost negligible, provided that $V(\boldsymbol{x})$ is already constructed for given power system modelling and $\boldsymbol{x}(t_e)$ is known;
- a single T-D simulation for the during-fault period to compute $\boldsymbol{x}(t_e)$; it is upper bounded by the CCT, i.e. it represents a very small percentage of the overall amount of simulations required by the T-D method with equivalent modelling;
- computation of $V_\ell = V(\boldsymbol{x}^u)$ which is almost negligible for a given $\boldsymbol{x}^u$; the problem is how to compute that $\boldsymbol{x}^u$ which provides a practically interesting limit value of V (see below).

Anticipated advantages and difficulties met. The above short description suggests that two main advantages were expected from the direct approach namely : straightforward computation of transient stability limits; suitable definition and computation of margins which in turn yields sensitivity analysis.

But on the other hand, in order to make the method applicable in practice, we should be able to :

(i) construct "good" $V-$functions and

(ii) assess "good" limit values or, equivalently, good practical stability domain estimates.

Both of the above conditions show to be extremely difficult to meet properly.

With reference to condition (i), it appeared quite early that constructing $V-$functions was only possible with over-simplified (actually unacceptably simplified) system modelling. Among the many interesting attempts to circumvent this difficulty, we mention the "structure preserving modelling" (Bergen and Hill, 1981) and the "pseudo-Lyapunov approaches" consisting of using T-D methods with detailed modelling together with pseudo-Lyapunov functions with simplified modelling (Athay *at al.*, 1979).

As for condition (ii), among the large variety of solutions proposed we mention the methods of Athay et al. (Athay *et al.*, 1979), of Kakimoto et al. (Kakimoto *et al.*, 1981), the acceleration approach (Ribbens-Pavella *et al.*, 1981), the BCU or exit point (Chiang *et al.*, 1994), the PEBS (Pai, 1989), and the IPEBS (Fonseca and Decker, 1985) methods. Nevertheless, although inventive, the various solutions did not show able to properly overcome the difficulties met. Indeed, on the one hand, the derived stability domain estimates were providing over-conservative stability assessment, with moreover unpredictably varying degree of conservativeness; on the other hand, from a computational point of view, the involved computations were often extraordinarily heavy, counterbalancing the computer gains expected of direct methods and even making them much heavier than T-D methods.

Resulting solutions : hybrid direct−T-D approaches. An impressively bulky research effort and impressive number of publications have been devoted to the derivation of (pseudo-) Lyapunov approaches, able to handle the transient stability problem in a way flexible (with respect to power system modelling), accurate (as compared to the corresponding T-D assessment) and computationally efficient. They eventually came up (explicitly or implicitly) with the following two observations :

- the stability domain estimation problem may be tackled by considering a two-machine or a one-machine equivalent of the multimachine power system
- the modelling problem may be solved by hybrid direct−T-D methods.

The first observation gave rise on the one hand to the Bellman decomposition-aggregation approach and vector Lyapunov functions, where the multimachine power system is decomposed into 2-machine subsystems (Grujić *et al.*, 1987)[4]. On the other hand to the single machine equivalent approaches (Rahimi and Schaffer, 1987; Xue *et al.*, 1988).

The second observation led to hybrid approaches, either of the multimachine type (Maria *et al.*, 1990), or of the single-machine equivalent type (Zhang *et al.*, 1997a).

The resulting practical approaches are hybrid direct−T-D methods of two types : the one uses the multimachine power system; the other a single machine equivalent (SIME) model[5].

The best performances achieved by these various ap-

[4]But their subsequent aggregation shows to impose extremely conservative stability conditions, unacceptable for practical purposes.

[5]Note that this latter relies on multimachine T-D simulations, at each time-step of which it computes the single-machine equivalent trajectory; its stability is then assessed on the basis of a direct criterion which allows reducing the length of required T-D simulations.

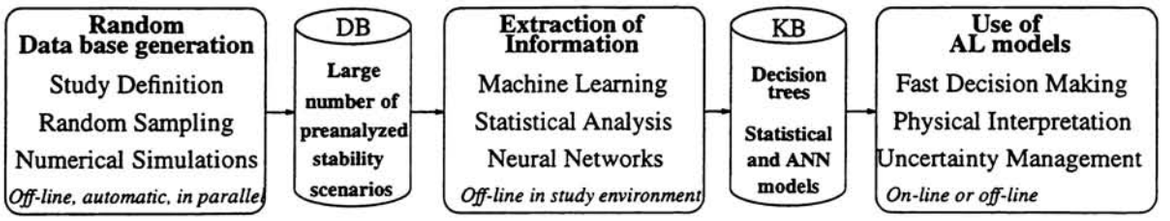


Fig. 1. AL framework for stability assessment. Adapted from (Wehenkel and Pavella, 1996)

proaches may be summarised as follows.

- Full flexibility with respect to power system modelling, as with T-D methods
- First-swing and multi-swing transient stability assessment, as with T-D methods
- Possibility of designing effective screening tools
- Computation of stability margins, yielding handy sensitivity analyses
- Identification of the relevant machines opening avenues towards real-time emergency control.

In short, the best of these methods appear to keep all the interesting properties of T-D methods while evading all their drawbacks and while being (5 to 20 times) faster than them. On the other hand, these methods circumvent all difficulties of pure Lyapunov methods at the expense of somehow (about 5 times) longer computations.

Concluding remarks. Over 30 years after their initiation, the investigations on direct methods applied to transient stability eventually reached and even surpassed the expectations initially put on them. Today, they have matured and are ready to use in the preventive mode : as a screening tool for planning, operation planning and real-time operation; for sensitivity analysis; for suggesting control actions. In the emergency mode for real-time transient stability emergency analysis and control, very promising results have been obtained with the "emergency SIME" approach (Zhang *et al.*, 1997b) which certainly deserves further research and development. It is anticipated that the emergency tools provided by this kind of approaches will start being implemented by the beginning of the coming century. This is a matter of communication facilities, available nowadays.

3.4 Automatic learning (AL) approaches

Here we deal with *supervised* learning only, for which we distinguish three families of approaches viz., machine learning - and the well-known subclass of decision trees (DTs); artificial neural networks (ANNs); statistical pattern recognition - and the well-known subclass of k nearest neighbours (kNNs).

Principle. All three families of approaches aim at extracting the essential information contained in a learning set of pre-analysed (i.e. pre-classified) cases. But while DTs and ANNs compress this information into synthetic rules or models (e.g. for classification), the kNNs learn in a case-by-case fashion, aiming at discovering similarities with the cases of the majority class of its k nearest neighbours (e.g. for classification). Figure 1 sketches the above general principle and suggests that the overall automatic learning process comprises in a sequence three steps : two off-line and one on-line.

Data base (DB) generation. The first step deals with the generation of the data base of pre-analysed cases. In the context of transient stability, these are states generated so as to cover all foreseen situations, using one (or a few number of) base-case(s) and random-though "realistic" variations of the parameters a priori likely to influence the transient stability phenomena. The stability of each one of these states with respect to a given disturbance is then assessed either in terms of margins (e.g. stability limits (CCT or power limit)) or simply as stable or unstable (with respect to a selected criterion; e.g. stable if its CCT is larger than a preassigned threshold value, unstable otherwise). The various types of stability assessment are obtained by running a T-D transient stability program with the desired degree of accuracy and power system modelling details.

The cases thus generated and pre-analysed are then subdivided in two independent sets : the learning set (LS) and the test set (TS). The LS will be used to learn the rules of the selected AL method, the TS to assess their quality.

The size of the DB, i.e. the number of cases deemed sufficient to design good rules depends on the problem at hand. Its choice is a matter of experience. To fix ideas, the DB described in (Wehenkel *et al.*, 1994) was generated for the purpose of transient stability assessment of an important power plant of the EDF extra-high voltage power system. It comprises 3000 states, 2000 used for the LS and 1000 for the TS. This DB allowed constructing "good" DTs and later on "good" ANNs, kNNs and fuzzy DTs (Wehenkel and Pavella, 1996).

The quality of the various methods' performances will be discussed below. The important observation we wish to convey here is that the quality of the information contained in the DB greatly depends on the knowledge of the human expert about his power system and its transient stability properties. Notice that since the DB is generated off-line, this expertise may carefully be exercised and, if necessary, augmented through suitable collaboration among experts. On the other hand, the quality of the resulting AL method strongly depends on the expertise of both the power system engineer and the AL method's designer. Note

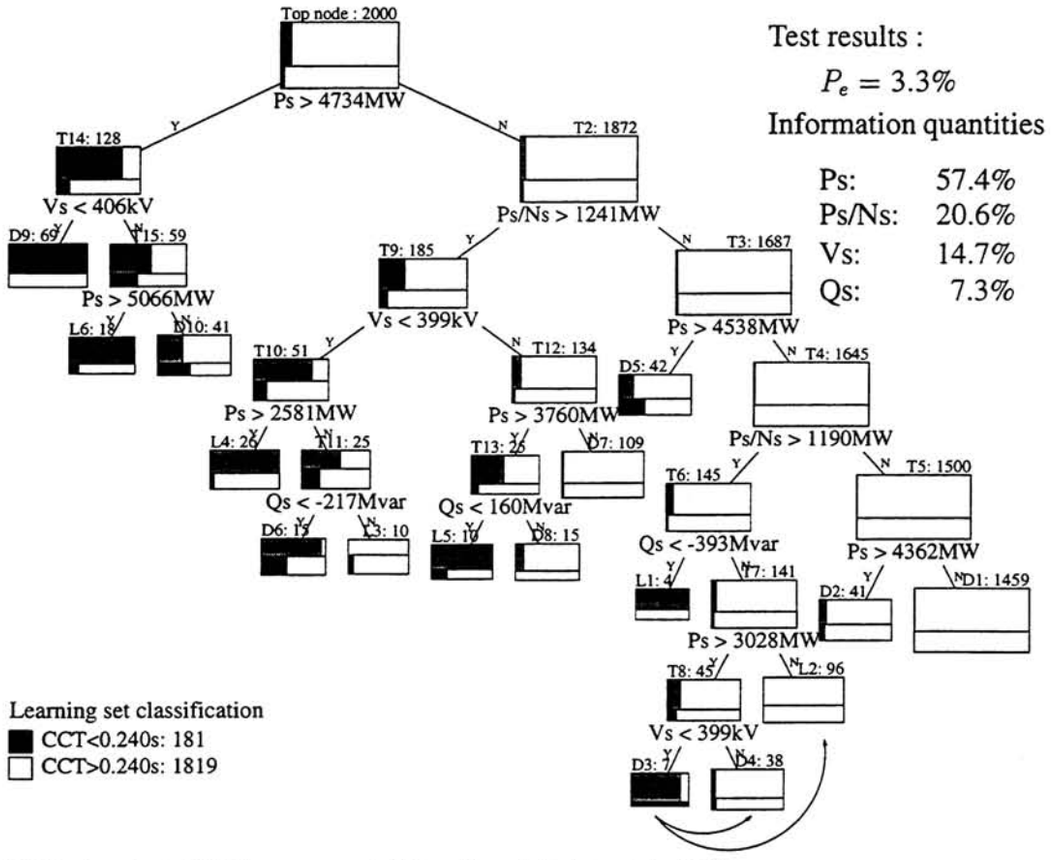


Fig. 2. DT for transient stability assessment. Taken from (Houben *et al.*, 1995)

also that further experience is gained while refreshing an already generated DB and the corresponding AL models.

Finally, we point out that a DB provides a statistical data analysis tool per se.

AL models and their use. Our purpose is essentially to identify the salient features of the various AL methods. However, elements of a method's design may help identify its features, in particular those of DTs shortly described below.

Decision trees. We illustrate our description on the DT portrayed in Fig. 2, obtained with the abovementioned DB. The tree comprises "test nodes" and "terminal nodes"; it involves dichotomic tests automatically carried out by the building procedure outlined below.

The tree building starts at the tree top node, with the entire learning set; this LS comprises a mixture of cases of the various classes. (In Fig. 2, two classes : stable and unstable). The procedure considers all "candidate attributes"[6] and chooses that attribute along with that threshold value of it which allows decomposing the mixture of stable-unstable cases into the two most purified learning subsets, directed towards the two successors of the node. This dichotomic procedure is repeatedly applied to each one of the successive test nodes, upon choosing the "best" test attribute along with its "best" threshold value. The procedure stops when getting a purified enough subset whose further splitting is deemed statistically meaningless. This subset composes a terminal node, labelled stable or unstable according to its majority population; for example, in Fig. 2, the leftmost terminal node D9 is an unstable node.

Let us outline the essential outcomes of the above procedure.

• Selection of the test attributes : these are a (generally small) subset of the candidate attributes proposed to the tree building. According to it, the test attributes are the parameters which drive the phenomena (here, transient stability). In Fig. 2, only 4 out of the 17 candidate attributes have thus been selected.

Besides identifying the test attributes, the tree also appraises their respective influence on the phenomena, in terms of their "information quantity" (IQ)[7]. Actually, IQ measures the purification ability of the corresponding attribute, i.e. its aptitude to best decompose the learning subset of the test node into its direct successors' subsets. Note that the IQ of a test attribute which appears more than once on the tree is the sum of its partial IQs. Note also that the higher the position of a test attribute in the tree hierarchy and the larger its IQ - and hence its influence on the phenomena.

[6] "Candidate attributes" are system parameters a priori deemed likely to drive the phenomena of concern. They are suggested by past experience and chosen by the human expert. Note that, for reasons appearing below, DTs built for the purpose of analysis in the preventive mode will advantageously use pre-disturbance system parameters.

[7] In Fig. 2, the test attributes' IQs are given in percent.

• Synthetic description of the physical phenomena of concern (here transient stability). The description follows the very hierarchical structure of the tree : the more essential the part of a test on the mechanism, the higher its position in the tree.

• Straightforward classification of (so far) unseen cases. This is achieved by placing the case at the top tree node and letting it progress down the tree by subjecting it to the successive tests that it thus meets. Eventually, the case reaches a terminal node and is given its class. To fix ideas, according to Fig. 2, a case whose P_s is larger than 4734 MW and whose V_s is larger than 406 kV will be directed to node D9 and classified unstable by the tree.

The former two outcomes refer to synthetic information which yields INTERPRETABILITY of the phenomena; the latter concerns detailed information and yields classification.

Actually, the tree unique feature and particular strength lies on the former : interpretability is essential - at least for stability concerns; for that reason, it will also be the backbone of the hybrid(ised) AL approaches advocated below. Another piece of information of great practical importance is the possibility to infer means to control. For example, Fig. 2 suggests how to enhance stability of a case directed to the lowest unstable terminal node (D3) : move it either to the next, stable terminal node (D4) by acting on the value of attribute V_s or to the immediately higher-level stable node (L2) by acting on the value of attribute V_s .

As concerning classification, the DT classifies a case merely relying on the values that the tree test attributes assume for that case. Hence, by choosing as candidate attributes pre-disturbance (and hence predetermined) system parameters, we reduce the classification computing effort merely to the time required by the case to traverse the tree, i.e. to an extremely short time (see below in Table 4). But, on the other hand, we should also observe that DTs provide a discrete (staircase) rather than smooth approximation. This prevents from getting back margin values (e.g. CCTs) which could have been used in the DB. In some sense, this discrete type of approximation is the price to pay in order to get good interpretability.[8]

To summarise the above discussion, in addition to interpretability, DTs provide means to :

- analysis (classification)
- sensitivity analysis (influence of the driving parameters on the phenomena)
- control

and that in extremely short times (fractions of ms, see in Table 4). In short, DTs provide all necessary ingredients to meet the needs identified in Section 2, and, in particular, needs for real-time operation.

Artificial neural networks. In contrast to DTs, ANNs are able to provide continuous margins (e.g. to recover the CCT information contained in the DB), at least in the neighbourhood of stability boundaries. This ability is owing to the nonlinear relationships between the ANN inputs and outputs, through their hidden layers' neurons. But, on the other hand, these complex relationships make it hardly possible to get an insight into the phenomena. Thus, for example, it is (almost) impossible to identify those input attributes which drive the phenomena and to appraise their influence. In some sense, in contrast to DTs, ANNs provide smooth approximations at the sacrifice of interpretability.

For our stability problems this "black-box" type of information is a handicap; it makes ANNs uninteresting for use, unless combined with other, more transparent approaches, like DTs.

k nearest neighbours. As already mentioned, kNNs learn in a case-by-case fashion. For example, a kNN classifies an unseen state in the majority class of its k nearest neighbours in the learning set. More generally, the ability to provide detailed information about a state by comparison with other, pre-analysed situations is an essential asset of kNNs. Another potentially very interesting property of kNNs is their ability to discover "outliers". These are cases "too far away" from all other cases of the DB; the kNN may warn about the possible failure of the derived AL method to assess them properly.

However, to make this conceptually attractive approach work effectively in practice, one has to resort to good optimisation techniques able to choose "good distances" among cases. In the context, "good distance" is meant to be good choice of the attribute space and of their attribute weights. Genetic algorithms can do the job; but they are very heavy. DTs can help quite effectively, as described below.

Fuzzy DTs. An interesting way to conciliate interpretability and smooth approximation is provided by fuzzy DTs, relying on fuzzy set theory. They are developed by choosing at each test-node a test attribute with two threshold values (instead of one for the crisp DTs) (Boyen and Wehenkel, 1995). This results in a DT with smooth (instead of staircase) approximation.

Although not yet mature enough, the fuzzy DT technique has at least three interesting properties namely : the possibility of recovering stability margins injected in the DB; a more accurate discrete classification; the possibility to perform sensitivity analysis (to compute derivatives of stability margins with respect to the test attributes).

Hybrid AL approaches. Each of the above "pure" AL approaches has its own assets and relative weaknesses. Hence the idea of combining them so as to gather their assets while evading their weaknesses. Of course, the choice among various combinations depends on the particular application sought.

Below we outline two such combinations developed for the purpose of transient stability and tested on the above-described DB. They both use DTs for their unique interpretability capabilities and for the informa-

[8] Although this is not fully true (see below the description of fuzzy trees.)

Table 4. Performance evaluation of AL methods
Taken from (Wehenkel and Pavella, 1996)

Method	P_e %		CPU time (SUN Sparc 20/50)	
	Classif.	Regr.	Off-line (s)	On-line (ms)
Crisp DT	3.3	–	60	0.1
Fuzzy DT	–	1.3	–	—
Pure ANN	–	1.6	33,000	3.6
Hybrid ANN	2.7	1.4	3,200	1.6
Pure kNN	6.6	6.7	200	100
Hybrid kNN	1.1	1.7	50,000	100

tion they provide in terms of test attributes, and their information quantity.

DT-ANNs : The DT provides a synthetic view of the phenomena and identifies the relevant parameters (its test attributes); these are then used as input variables to the ANN. The ANN may output either continuous type of margins or discrete classification, more accurate than that of the DT.

DT-kNNs : The DT provides a synthetic view of the phenomena, including the test attributes along with their IQ. The kNN uses these attributes to define the attribute space and their respective IQs to weight them in the distance measure. The DT-kNN model may further be optimised using genetic algorithms, as proposed in (Houben *et al.*, 1997).

Two main outcomes of the DT-kNN combination are : description of the global mechanism of the phenomena by the DT; description of the detailed mechanism concerning an (unseen) case by comparison with its (known) nearest neighbours by the kNN. This latter detailed information may be used in various ways, as for example in order to suggest means to control so as to reinforce a case's stability.

Other interesting outcomes are the identification of outliers and the avoidance of dangerous errors, i.e. of dangerous misclassifications (Houben *et al.*, 1997); such a dangerous error arises when an actually (very) unstable case is declared stable by the classifier.

Overall comparison. Tables 4 and 5, borrowed from (Wehenkel and Pavella, 1996), give a synoptic view of the performances and salient features of the above considered "pure" and "hybrid" AL approaches. Table 5 gathers general comparative properties. On the other hand, Table 4 provides the particular performances of the AL methods applied to the transient stability problem and the DB described previously. Let us recall that actually 2000 cases of this DB were used to learn the various AL methods, the other 1000 cases were used to assess their accuracy, or their corresponding test set error rate, P_e .

Observe that the best accuracy performance for classification is here provided by the hybrid kNN (last row of Table 4). Actually, this is a kNN-DT-GA model where the genetic algorithm (GA) allows enhancing the accuracy of kNN-DT) at the expense of heavy computations.

Table 5. Salient features of AL methods applied to TSA
Taken from (Wehenkel and Pavella, 1996)

	Method	Functionalities	Computational	
			Off-line	On-line
Pure	Crisp DTs	**Good interpretability (global).** Discrete. Good accuracy for simple "localised" problems. Low accuracy for complex, diffuse problems.	Very fast	Very fast
Pure	ANNs	**Good accuracy.** Low interpretability. Possibility for margins and sensitivities.	Very slow	Fast
Pure	kNN	**Good interpretability (local).** Conceptual simplicity.	Very slow	Very slow
Hybrid	Fuzzy DTs	**Good interpretability (global).** Symbolic and continuous. More accurate than crisp trees. Possibility for margins and sensitivities.	Slow	Fast
Hybrid	DT-ANN	Combine features of DTs and ANNs	Slow	Fast
Hybrid	DT-kNN	Combine features of DTs and kNNs	Slow	Slow

The three main intrinsic assets of AL methods. The foregoing discussions identify and Table 5 highlights three salient assets intrinsic to (some of the) AL methods over the deterministic ones, namely :

- interpretability of the phenomena;
- extraordinary on-line computational efficiency;
- management of uncertainties.

The former two properties are linked to the very AL methods' models. Management of uncertainties is obtained thanks to the statistical nature of AL methods, i.e. to the very way of generating a DB, where the uncertain information is (suitably) randomized. This latter property is of great importance in real-world problems where uncertainties are always present. They are inherent to the necessarily limited knowledge of the power system modelling (e.g. modelling of loads connected to high voltage buses) and parameters, not to mention the lack of information due to ecomonical-political factors.

Application of AL methods to other stability problems. So far, the AL methodology was outlined in the context of transient stability. Actually, this same methodology may readily be adapted to the specifics of other power system stability problems, as those described in Table 1. The adaptation essentially concerns the DB generation. Such developments have already been achieved in the context of voltage stability (Wehenkel *et al.*, 1991) and steady-state security (Hatziargyriou *et al.*, 1994).

Real-world applicability concerns. The above description shows that AL methods are indeed able to meet very stringent needs of power system stability assessment. However, their actual implemen- tation and use should overcome at least two essential difficulties. The one is linked to the DB generation, which is the prerequisite of all AL methods : to make it a routine task, suit-

Table 6. Transient stability assessment : techniques vs criteria

	Criteria	1	2		3	4	5					
			Inform. req.		Off-line	Real-time	Information provided					
		Model. poss.	Off-line	Real-time	prep. tasks	comput. reqrts	Anal.	Screen.	Interpret.	Sensit.	Control	Managt of uncert.
Time-domain		+++	FD	CS	mV	B	D		–	–	–	–
Autom. Learning methods	Off-line prepar.	+++	FD		DB , V		D , S		+	+		+
	Real-time use	+++		SV		EL	D , S		+	+	P , C	+
Direct methods	"Pure"	+,++	PD	CS	V	L	D	++	–	+	P	–
	"Hybrid" preventive	+++	FD	CS	mV	FB	D , S	+	–	+	P	–
	"Hybrid" emergency	+++	–	SV	mV	EL	S		+	+	C	+

FD : Fully Detailed; PD : Partly Detailed; CS : Complete System picture; SV : Selected Variables; D : Detailed; S : Synthetic; P : Preventive; C : Corrective; DB : Data Base; V : Validation; mV : minor Validation; B : Bulky computations; FB : Fairly Bulky computations; L : Light computations; EL : Extremely Light computations

able tools must become available[9]. The other reason is that all strategies developed so far rely on deterministic concepts : in order for AL methods to be accepted by the power system community, new, different types of strategies should be devised. Thus, it is very likely that AL methods will at first be used in areas where they prove to be superior to deterministic approaches; such a priority area seems to be real-time operation (in the preventive mode, see last row of Table 2, and/or in the emergency mode).

Maintaining expertise in off-line study environments is yet another way of taking due advantage of statistical approaches.

Towards a unified approach to power system stability myth or anticipated reality ?. While decomposition of power system stability into many sub-problems is a convenient way to tackle such intricate issues, the overall stability of the system should always be kept in mind : solutions of a stability sub-problem should not spoil another.

Ref. (Wehenkel *et al.*, 1997) proposes a global probabilistic approach to the dynamic performance of a power system subjected to various severe fast and slow dynamic phenomena. The approach models in a probabilistic way potential causes leading to extreme consequences (such as voltage collapse, overloaded line trippings or loss of synchronism), while taking into account modelling uncertainties. The DB is generated by randomly combining the various causes. It is then analysed using statistical tools.

Such a DB has been generated for a part of the EDF power system, using about 1,500 simulation scenarios and a simulation model with over 11,000 state variables.

To handle such bulky data, the authors have been led to develop special-purpose tools, namely : (i) the DB generation tool, exploiting parallel computations to simulate in detail the scenarios; (ii) a statistical data mining tool especially adapted to very large DBs composed of temporal data.

The first obtained results are very encouraging. Thus, although much work is still needed before reaching the ultimate goal of synthetic stability analysis and control, we may confidently anticipate that the approach will eventually become reality.

Concluding remarks. The tremendous progress in computer technology has been determinant to the development of AL methods. In the context of transient stability (and to a lesser extent of voltage stability), the achievements of these methods in less than 10 years is impressive. A major factor to this success is due to the close collaboration and involvement of R&D engineers, who provided their expertise and critical judgement. They have contributed significantly to generate good DBs, which are the keystone of AL methods.

The next - and final step towards implementation is to make AL methods be accepted by practicing engineers and personel in charge of power systems operation. This is now under good way.

4. COMPARISON OF METHODS

Section 2 has identified present needs of transient stability assessment in the context of both preventive and emergency modes.

Section 3 has described the three classes of existing approaches to transient stability assessment along with their couplings, highlighted their potentials and attempted to identify the needs that each one of them is able to meet. It was suggested that some of the needs may be met by more than one method.

Another type of comparison is proposed in Table 6, where the various techniques are assessed in terms of the following five evaluation criteria : 1) Modelling possibilities; 2) Type of information required; 3) Off-line preparation tasks; 4) Real-time computational requirements; 5) Type of information provided.

We should specify that this table takes into consideration the best performances of each class of methods.

Finally, let us mention that, with minor changes, the

[9] Actually, such tools for the systematic DB generation have already been developed in the context of operation planning (Jacquemart et al., 1995); also, in the context of real-time, an authomatic operator is beeing proposed by Dy-Liacco (Dy-Liacco, 1996).

first two rows of the table (relative to comparisons of T-D and AL methods) could be extended to voltage stability assessment.

5. CONCLUSION

This paper has attempted to identify the various classes of methods nowadays available for transient stability and more generally for stability of power systems. It has shown that stability assessment tasks are much more demanding now than by the past. At the same time the methods have significantly progressed; they thus became able to handle problems whose solution was considered to be out of reach only a few years ago.

However, enthusiastic claims in anticipation of successful achievements may have caused reluctant reactions against innovative methods. The author would therefore like to convey the following threefold observation :

- nowadays techniques do exist, which are able to meet the most stringent needs of each type of stability;
- because of the high degree of sophistication of (some of) these techniques, their actual implementation and use sorely need the close collaboration between researchers, developers and practicing engineers in the field;
- the final goal should attempt to imbed all types of stability assessment through a unified approach instead of piecewise approaches. This goal has not yet been reached; but the problem is being identified and properly stated : this could be half-way from its solution.

Abbreviations and acronyms

All abbreviations are fully defined at the place where they are first introduced. As a convenience to the reader we have collected below some of the more frequently used.

AL	:	automatic learning
ANN	:	Artificial neural network
CCT	:	critical clearing time
DB	:	data base
DT	:	decision tree
kNN	:	k nearest neighbour
T-D	:	time-domain
TSA	:	transient stability assessment

REFERENCES

Athay T., V.R. Sherkat, R. Podmore, S. Virmani and C. Puech (1979). Transient energy stability analysis. In *System Engineering for Power : Emergency Operating State Control - Section IV.*, U.S. Dept of Energy Publication, No. CONF-790904-P1.

Bergen A.R. and D.J. Hill (1981). A structure preserving model for power system stability analysis. In *IEEE Trans. on PAS*, PAS-100, pp. 25-33.

Boyen, X. and L. Wehenkel (1995). Fuzzy decision tree induction for power system security assessment. In *2nd IFAC Symp. on Control of Power Plants and Power Systems*, Cancun, Mexico, December, pp. 151-156.

Chiang H.D., F.F. Wu and P.P. Varaiya (1994). A BCU method for direct analysis of power system transient stability. In *IEEE Trans.*, PWRS-9, pp. 1194-2000.

Dy-Liacco T.E. (1968). *Control of Power Systems via the Multi-Level Concept.* Sys. Res. Center, Case Western Reserve Univ., Cleveland, Ohio, Rep. SRC-68-19.

Dy-Liacco T.E. (1996). On the applicability of automatic learning to power system operation. In *Revue E.* Special issue on *Apprentissage Automatique; Applications aux Reseaux d'Energie Electrique*, December.

Fonseca L.G.S. and I.C. Decker (1985). Iterative algorithm for critical energy determination in transient stability of power systems. In *IFAC Symp. on Planning and Operation Electric Energy Systems*, Rio de Janeiro, Brazil.

Fouad A.A. and V. Vittal (1992). *Power System Transient Stability Analysis Using the Transient Energy Function Method.* Prentice-Hall Inc., Englewood Cliffs, N.J.

Grujić Lj.T., A.A. Martynyuk and M. Ribbens-Pavella (1987). *Large Scale Systems Stability Under Structural and Singular Perturbations.* Lecture Notes in Control and Information Sciences Series, Springer-Verlag, Berlin.

Hatziargyriou N.D., G.C. Contaxis and N.C. Sideris (1994). A decision tree method for on-line steady state security assessment. In *IEEE Trans.*, PWRS-9, pp. 1052-1061.

Houben I., L. Wehenkel and M. Pavella (1995). Coupling of k-NN with decition trees for power system trnsient stability assessment. In *Proc. of the IEEE/CCA'95 Conf. on Control Applications*, Albany, September, pp. 825-832.

Houben I., L. Wehenkel and M. Pavella (1997). Genetic algorithm based k nearest neighbours. In *Proc. of Conf. on Control of Industrial Applications*, Belfort, France, May (to appear).

Jacquemart Y., L. Wehenkel, T. Van Cutsem and P. Pruvot (1995). Statistical approaches to dynamic security assessment : The data base generation problem. In *Proc. of SIPOWER'95, 2nd IFAC Symp. on Control of Power Plants and Power Systems*, December, pp. 243-246.

Kakimoto N., Y. Ohsawa and M. Hayashi (1980). Transient stability analysis of multimachine power systems with field flux decays via Lyapunov's direct method. In *IEEE Trans.*, PAS-99, pp. 1819-1827.

Kimbark E.W. (1948). *Power System Stability. Vol.I : Elements of Stability Calculations.* John Wiley and Sons, Inc. New York.

Kundur P., E.G. Neudorf, P. Henry, J. Fish and J. Mitsche. (1987). Rapid analysis of transient stability : needs, benefits and requirements. In *Rapid Analysis of Transient Stability* (sponsored by System Dynamic Performance Subcommittee of the IEEE

Power Systems Eng. Committee of the IEEE-PES; W.W. Price, Chairman of the Task Force), *IEEE/PES Winter Meeting*, Paper No. 87THO 169-3-PWR.

Kundur P. (1994). *Power System Stability and Control.* McGraw Hill.

Maria G.A., C. Tang and J. Kim. (1990). Hybrid transient stability analysis. In *IEEE Trans.*, PWRS-5, pp. 384-394.

Pai, M.A. (1981). *Power System Stability. Analysis by the Direct Method of Lyapunov.* North-Holland Publishing Company, Amsterdam.

Pai, M.A. (1989). *Energy Function Analysis for Power system Stability.* Kluwer Academic Publishers, Boston.

Park R.H. (1929). System stability as a design problem. In *AIEE Trans.*, 48, pp. 170-194.

Pavella M. and P.G. Murthy (1993). *Transient Stability of Power Systems.* J. Wiley and Sons, Chichester, England.

Prabhakara F.S. and A.H. El-Abiad (1975). A simplified determination of transient stability regions for Lyapunov methods. In *IEEE Trans*, PAS-94, pp. 672-689.

Prabhakara F.S. and G.T. Heydt (1987). Review of pattern recognition methods for rapid analysis of transient stability assessment. *IEEE PES Winter Meeting*, New York, Paper No. 87THO 169-3 PWR

Rahimi F.A. and G. Schaffer (1987). Power system transient stability indexes for on-line analysis of 'worst case' dynamic contingencies. In *IEEE Trans.*, PWRS-2, pp. 660-668.

Ribbens-Pavella M. P.G. Murthy and J.L. Horward (1981). An acceleration approach to practical stability domain estimation in power systems. In *Electrical and Power System Modelling and Simulation* (J. Robbert and W. Midvidy, Eds), J.C. Baltwer Ag, Scientific Publishing Co, pp. 3-10.

Sobajic, D.J. and Y.H. Pao (1989). Artificial neural-net based dynamic security assessment for electric power systems. In *IEEE Trans. on PWRS*, Vol.4, No.1, February, pp. 220-228.

Wehenkel L., Th. Van Cutsem and M. Ribbens-Pavella (1989). An artificial intelligence framework for on-line transient stability assessment of electric power systems. In *IEEE Trans.*, PWRS-4, pp. 789-800.

Wehenkel L., Th. Van Cutsem, M. Gilliard, M. Pavella, B. Heilbronn and M. Goubin (1991). Decision trees for preventive voltage stability assessment. In *Procs of the Int. Workshop on Bulk Power System Voltage Phenomena - Voltage Stability and Security*, Deep Creek Lake, McHenry, Maryland, pp. 217-228.

Wehenkel L., M. Pavella, E. Euxibie and B. Heilbronn (1994). Decision tree based transient stability method. A case study. In *IEEE Trans.*, PWRS6-9, pp. 459-469.

Wehenkel L., and M. Pavella (1996). Why and which automatic leraning approaches to power system security assessment. In *Proc. of the IMACS-IEEE Conf. on Computational Engg. in Systems Applications*, Lille, France, July,

Wehenkel L., C. Lebrevelec, M. Trotignon and J. Batut (1997). A probabilistic approach to the design of power systems protection schemes against blackouts. *Proc. of IFAC/CIGRE Symp. on Control of Power Systems and Power Plants*, Beijing, China, August (submitted).

Xue Y., Th. Van Cutsem and M. Ribbens-Pavella (1988). A simple direct method for fast transient stability assessment of large power systems. In *IEEE Trans.*, PWRS-3, pp, 400-412.

Zhang, Y., L. Wehenkel, P. Rousseaux and M. Pavella (1997a). SIME : a hybrid approach to fast transient stability assessment and contingency selection. In *Int. Journal of Electrical Power and Energy Systems*, Vol.19, No.3, March, pp. 195-208.

Zhang, Y., L. Wehenkel, and M. Pavella (1997b). An approach to real-time transient stability emergency control. In *Proc. of IFAC/CIGRE Symp. on Control of Power System and Power Plants*, Beijing, China, August (to appear).



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Transient Stability of Power Systems. Theory and Practice ;
M. Pavella, P.G. Murthy. J. Wiley & Sons, London, 1993, 403 pages

Trends on real-time transient stability assessment of electric power systems ;
Electrical and Power systems Modelling and Simulation, J.B. Baltzer AG, Scientific Publishing Co. IMACS 1989, pp. 3-10.

Dynamic state estimation techniques for large-scale electric power systems ;
P. Rousseaux, M. Pavella. Control and Dynamic Systems, Vol.41, Academic Press, 1991, pp.79-133.